UCLouvain

Combinatorial optimization
Homework
Report

Optimal design of an Immunization Campaign

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Table des matières

1 Modeling the problem **3**
1.1 Variables . . . 3
1.2 Objective and constraints . . . 4
1.2.1 Constraints . . . 4
1.2.2 Objective function . . . 6

2 Relaxation of the problem **6**

3 Improving the model by valid inequalities **6**

4 LP relaxation **7**

5 A first primal bound **7**
5.1 Some properties . . . 7
5.2 Greedy approach . . . 7
5.3 Local search . . . 8

6 Implementation and numerical experiments **9**
6.1 Implementation . . . 9
6.1.1 coord20-5-2_processed.dat . . . 10
6.1.2 coord50-5-1_processed.dat . . . 11
6.1.3 coord100-10-2b_processed.dat . . . 11
6.2 Numerical experiments . . . 12
6.2.1 Greedy Algorithm *VS* Greedy + Local Search . . . 13
6.2.2 OptVax1 *VS* OptVax2 . . . 13
6.2.3 OptVax1 *VS* OptVax2 with warm-start . . . 13
6.2.4 OptVax2-LP . . . 14
6.2.5 General comparison . . . 14

7 Primal bound with an advanced problem specific heuristic **16**
7.1 Movements . . . 16
7.2 Tabu Search . . . 17
7.2.1 Algorithm . . . 17
7.2.2 Example . . . 17
7.2.3 Comparison . . . 18

8 Dual bound via Lagrangian Duality **20**
8.1 Complicating constraints . . . 20
8.1.1 Constraints Similar to a Knapsack Problem . . . 20
8.1.2 Constraints Similar to an Assignment Problem . . . 21
8.1.3 Constraints Similar to a Shortest Path Problem . . . 21
8.1.4 Reformulation of the objective function . . . 22
8.2 Sub-gradient Descent . . . 22
8.3 Step size strategy . . . 22
8.3.1 Influence of λ_0 . . . 22
8.3.2 Influence of a in $h_k = \frac{a}{k+b}$ . . . 23
8.3.3 Influence of b in $h_k = \frac{a}{k+b}$ . . . 24

9 Numerical Experiments **24**

10 Conclusion **25**

Part 1 - Model, solve and relax !

1 Modeling the problem

1.1 Variables

Here we give all variables we will use in the model.

1. Let $x_i \in \{0,1\}$ for $i \in I$ be the variable that decides if the VC i should be open or not.
$$x_i = \begin{cases} 1 & \text{if VC } i \in I \text{ is opened,} \\ 0 & \text{otherwise.} \end{cases}$$

2. Let $y_{ij} \in \{0,1\}$ for $i \in I$ and $j \in J$ be the variable that indicates whether locality j is covered by VC i or not.
$$y_{ij} = \begin{cases} 1 & \text{if the VC } i \in I \text{ is opened and cover the locality } j \in J, \\ 0 & \text{otherwise.} \end{cases}$$

3. Let $v_m \in \{0,1\}$ for $m \in \{1,...,M\}$ be the variable that decides if the MMT m leaves the chosen VC or not (in order to simplify the notation, we define the set $M := \{1,...,M\}$).
$$v_m = \begin{cases} 1 & \text{if MMT } m \in M \text{ is used,} \\ 0 & \text{otherwise.} \end{cases}$$

4. Let $t_{mj} \in \{0,1\}$ for $m \in M$ and $j \in J$ be the variable that indicates whether the locality j is visited by MMT m or not.
$$t_{mj} = \begin{cases} 1 & \text{if the MMT } m \in M \text{ is used and cover the locality } j \in J, \\ 0 & \text{otherwise.} \end{cases}$$

5. Let $w_{mjk} \in \{0,1\}$ for $m \in M$ and $j,k \in I \cup J$ be the variable that decides if the MMT m goes from the location j to location k or not.
$$w_{mjk} = \begin{cases} 1 & \text{if the MMT } m \in M \text{ goes from locality } j \in I \cup J \text{ to } k \in I \cup J, \\ 0 & \text{otherwise.} \end{cases}$$

6. Let $u_{mj} \in \mathbb{N}$ for $m \in M$ and $j \in J$ be the variable that indicates in which position the locality j is on the route of the MMT m.
$$u_{mj} = \begin{cases} n & \text{if the MMT } m \in M \text{ visits the locality } j \in J \text{ in n-th position,} \\ 0 & \text{otherwise.} \end{cases}$$

1.2 Objective and constraints

1.2.1 Constraints

1. One and only one vaccination center must be opened :

$$\sum_{i \in I} x_i = 1.$$

2. A locality can be either ; not visited, visited at most once by a MMT or within the range of an opened VC :

$$\forall j \in J \qquad \sum_{m \in M} t_{mj} + \sum_{i \in I} y_{ij} \leq 1.$$

Explanation : if the left term is equal to 1, it means that the locality is visited by a MMT. In the case where the right term is equal to 1, it indicates that the locality is covered by a MMT. As both term could be 0, the locality may be not visited.

3. Each route corresponds to a mobilized MMT :

$$\forall j \in J \qquad \sum_{m \in M} \sum_{\substack{k \in I \cup J \\ k \neq j}} w_{mjk} \leq 1$$

$$\forall k \in J \qquad \sum_{m \in M} \sum_{\substack{j \in I \cup J \\ j \neq k}} w_{mjk} \leq 1.$$

Explanation : these two equations ensure that the sum of all MMT arriving to or starting from a locality must be at most 1.

$$\forall m \in M, \quad \forall k \in I \cup J \qquad w_{mkk} = 0.$$

Explanation : The constraint forces the MMT to move, i.e. a MMT can't goes to location k to location k.

$$\forall m \in M, \quad \forall k \in I \cup J \qquad \sum_{\substack{j \in I \cup J \\ j \neq k}} w_{mjk} = \sum_{\substack{j \in I \cup J \\ j \neq k}} w_{mkj}.$$

Explanation : the sum of all MMT starting from a location is exactly the sum of MMT arriving to this location.

$$\forall m \in M, \quad \forall j, k \in J, k \neq j \qquad w_{mjk} + u_{mj} \leq u_{mk} + (|I| + |J|)(1 - w_{mjk}).$$

Explanation : this is the sub-turns constraint. It ensures that if a MMT goes from locality j to locality k, the position of k in the route of the MMT m is one more than that of j.

$$\forall m \in M, \quad \forall j \in J \qquad \sum_{k \in I \cup J} w_{mkj} = t_{mj}.$$

Explanation : this constraint ensures that the variable t_{mj} does what we want, i.e. a locality j is visited by a MMT if there is one that arrive at this locality.

4. Each route must respect the capacity constraint :

$$\forall m \in M \qquad \sum_{j \in J} t_{mj} q_j \leq v_m Q.$$

Explanation : the sum over the citizens that a MMT visits, must be lower or equal to the capacity of the vehicle. If this MMT doesn't leave the chosen VC, then the sum must be equal to 0.

5. The capacity for the VC is only for specifically conditioned doses delivered by MMTs :

$$\forall i \in I \qquad x_i \sum_{m \in M} \sum_{j \in J} t_{mj} q_j \leq C_i.$$

Explanation : the sum over the citizens visited by all MMTs must be lower or equal to the capacity constraint of the chosen VC. Notice that the factor x_i ensure that we are just taking the chosen VC into account, indeed, for an other VC the constraint will always be valid.

6. The starting point and the end point are the same for each route and turns out to be the chosen VC :

$$\forall i \in I \qquad \sum_{m \in M} \sum_{j \in I \cup J} w_{mij} = x_i \sum_{m \in M} v_m$$
$$\forall i \in I \qquad \sum_{m \in M} \sum_{j \in I \cup J} w_{mji} = x_i \sum_{m \in M} v_m.$$

Explanation : these two constraints ensure that the sum of MMTs arriving/leaving the chosen VC is exactly the sum of all used MMTs. The factor x_i forces to not have MMT arriving/leaving an other VC.

7. All prioritized localities have to be covered :

$$\forall j \in \hat{J} \qquad \sum_{m \in M} t_{mj} + \sum_{i \in I} y_{ij} = 1.$$

Explanation : forces all prioritized localities to be either visited by a MMT or covered by the chosen VC.

8. *Additional constraints.*

- Budget constraint :

$$\sum_{i \in I} x_i f_i + \sum_{m \in M} v_m p + \sum_{m \in M} \sum_{k \in I \cup J} \sum_{j \in I \cup J} d_{ij} w_{mjk} \leq B$$

Explanation : for the budget constraint, we have to sum the cost of opening the chosen VC, the cost of all routes opened and the cost of distance travelled.

- Ensuring that the variable y_{ij} does what we want :

$$\forall i \in I, \quad \forall j \in J \qquad y_{ij}(d_{ij} - r_i)x_i \leq 0 \tag{1}$$
$$\forall i \in I, \quad \forall j \in J \qquad (1 - y_{ij})(r_i - d_{ij})x_i \leq 0 \tag{2}$$
$$\forall i \in I, \quad \forall j \in J \qquad y_{ij}(1 - x_i) = 0. \tag{3}$$

Explanation : first notice that the factor x_i in (1) and (2) implies that we are just taking the chosen VC into account, since for the other, the inequalities will always be valid. Thus (1) ensures y_{ij} to be 0 if $r_i < d_{ij}$, meaning that locality j is not in the radius of the VC. (2) does the same for the inverse case. Finally, (3) forces all y_{ij} to 0 if the VC i is not chosen.

1.2.2 Objective function

Since we want to optimize the total number of citizens vaccinated, we will have the following objective function :

$$\text{Maximize } \sum_{j \in J} q_j \left(\sum_{m \in M} t_{mj} + \sum_{i \in I} y_{ij} \right).$$

2 Relaxation of the problem

By fixing the location of the Vaccination Center and assigning a subset of localities to each Mobile Medical Team (MMT), the problem reduces to a form of Travelling Salesman Problem (TSP). For each MMT, we need to find the optimal route that visits the assigned localities.

Therefore, the relaxed problem can be understood as a multi-vehicle TSP with a fixed point of departure and arrival (the VC). Each MMT's route is a solution to a TSP, and the objective is to minimize the total travel cost.

In cases, where there is no subset of localities assigned to the MMTs, the final objective of the problem is to maximize the quantity q of vaccinated people (respecting the budget, and the other constraints). Therefore, some routes may not be the shortest possible, as they will prioritize maximizing q instead. As we shall see later, if we adjust the budget, particularly by introducing C_{km}, a travel cost per kilometre greater than 1, we can force the routes to be shorter and still keep the objective of maximizing the quantity q of vaccinated people.

The constraint matrix in this relaxed problem is not totally unimodular. This problem still involves capacity/budget constraints, and can't be composed of just {-1,0,1} values, which is a necessary condition to be TU.

3 Improving the model by valid inequalities

We want a lower bound on the number of vehicles to be used, the idea we will express is that Q times the number of MMTs must be greater or equals to the total number of vaccinated people. Obviously, if this was smaller, this means that at least one MMT exceed his capacity constraint. We thus have the following additional constraint :

$$\sum_{m \in M} v_m \geq \frac{1}{Q} \sum_{j \in J} q_j \left(\sum_{m \in M} t_{mj} \right).$$

Hence the inequality gives us a lower bound for the number of MMTs deployed. Theoretically, adding this constraint won't change the optimal solution of the model. Still, it is not useless. Adding constraint permit to reduce the feasible space of our model, especially for the relaxation of the problem (see next section), and thus sometimes, it helps the solver to find the best solution.

4 LP relaxation

The LP relaxation is a technique use to find a upper bound on our problem. As it involves the suppression of the integer constraints of all variable, it may permit the solver to improve the optimal solution. In our case, deploying half a car becomes a feasible solution for example.

The changes are therefore only on the variables which becomes :

- $x_i \in [0, 1]$ for $i \in I$;
- $y_{ij} \in [0, 1]$ for $i \in I$ and $j \in J$;
- $v_m \in [0, 1]$ for $m \in M$;
- $w_{mjk} \in [0, 1]$ for $m \in M$ and $j, k \in I \cup J$;
- $u_{mj} \in \mathbb{R}$ for $m \in M$ and $j \in J$;

where each one is define the same than in the subsection *variables*.

5 A first primal bound

5.1 Some properties

Let X be the vector stacking all the decision variables and $z(X)$ the objective function .

Let $z^* = \max_{X \in F} z(X)$, the optimal value of the problem. Define,

$$z_{max}^{FVC}(X) = \max_{i \in I} \{z(X) \mid \text{ s.t. VC is opened at location } i \text{ in } X\}$$

We want to find a necessary and sufficient condition on X such that the following equation holds :

$$z_{max}^{FVC}(X) = z^*$$

The condition is that there exists a VC i, and a feasible solution X^* where $z(X^*) = z^*$. This solution is necessarily optimal compared to the other solutions obtained for the other VCs. We will then have $z_{max}^{FVC}(X) = \max_{i \in I} \{z(X)\} = z(X^*)$ and so $z_{max}^{FVC}(X) = z^*$

5.2 Greedy approach

If we suppose that one (and only one) VC is fixed. The greedy algorithm for this problem consists to create a first MMT that starts from this fixed VC, and visits the nearest locality that is not within the VC's range. After visiting this first locality, the MMT go to the next nearest locality, continuing in this way until no more localities can be visited without violating the constraints. Indeed, before visiting the next localities, we check if

it respects the constraints of capacity Q, , the capacity $C[VC_{chosen}]$, and the budget B. For the budget constraint, we need to verify that there is enough remaining budget to travel to the next locality and return to the VC, in the case the next locality is the last one visited by the current MMT. When these constraints are not respect for the current MMT, if $B_{remaining} > p$ and the constraint of maximum M MMTs is still respected, we open a new MMT, proceed as before, and continue until we are blocked by one or more constraints.

Once the greedy solution is constructed, we verify if all high-priority localities (that are not in the VC's range) have been visited. If any high-priority locality is missing, we restart at 0 with now a focus on visiting the high-priority localities first.

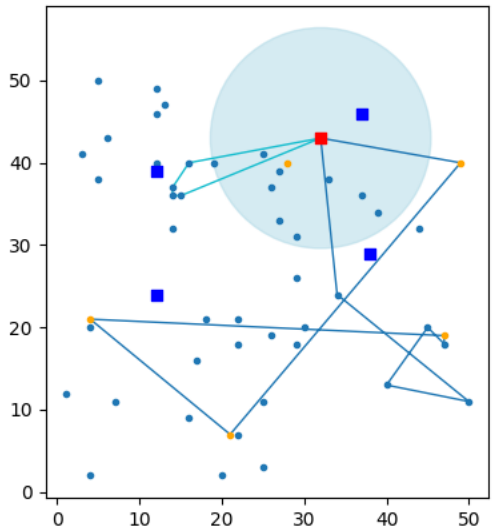
If no prior information is available about the fixed VC, we can extend the greedy approach by considering all possible VC. For each VC, we apply the greedy approach and the best solution (i.e. the solution where the most people are vaccinated) across all possible VC is then selected. This greedy approach provides a primal bound for the problem.

5.3 Local search

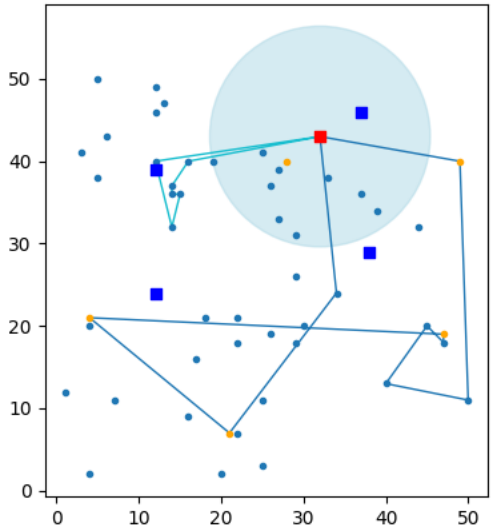
The local search algorithm starts with the greedy solution and iteratively improves it by exploring neighbour solutions.

The local search algorithm starts by calculating the total distance of the given route obtained after the greedy approach. For each route, the algorithm tries to improve the solution by reversing segments of the route and recalculating the total distance. If this change gives a shorter distance of the route, the new route is kept as the best solution. This process is repeated for all routes, and the algorithm returns a list of the improved routes.

After this search, we return to the greedy approach, where we keep the improved routes (except the last one) and try to improve the last one. Indeed, the last route constructed with the greedy algorithm is often constrained by budget (or the capacity C), unlike the other routes, which are constrained by the capacity Q. Thus, the budget saved by the shorter distance of the routes from $[0, m-1]$ can sometimes allow the last route to be extended.



Greedy approach



Local search amelioration

The example above is inspire from coord50-5-3b_processed.dat, **but with changes of the parameters $C_{\mathbf{km}}$ (cost per kilometer greater than 1) to provide a simple and clear illustration**. Thanks to the local search, the MMTs can visit an other locality, increasing the total number of vaccinated people from 362 to 389.

6 Implementation and numerical experiments

6.1 Implementation

We have realized an implementation of :

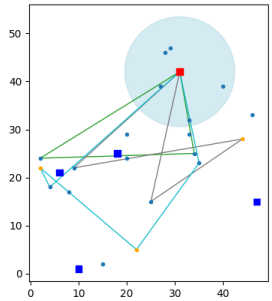
- OptVax1
- OptVax2
- OptVax2-LP
- Greedy algorithm and local search

As well as a solution visualizer (The open VC is in red, its range is represented by a blue disk, and the prioritized localities are in orange).

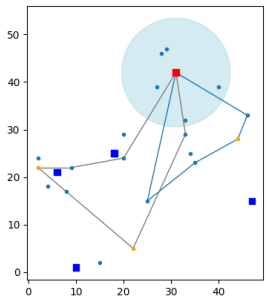
As explained before, the objective is to maximize the number of people that can be fully vaccinated. For the illustration we have sometimes reduced the budget, by introducing C_{km}, a travel cost per kilometre greater than 1, to force the routes to be shorter just for a better visualisation (ensuring that the budget did not decrease enough to impact the number of people vaccinated).

Here are some examples of the results obtained :

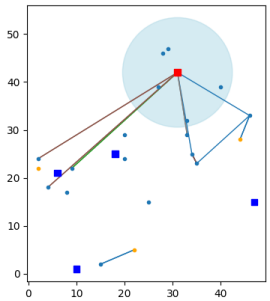
6.1.1 coord20-5-2_processed.dat



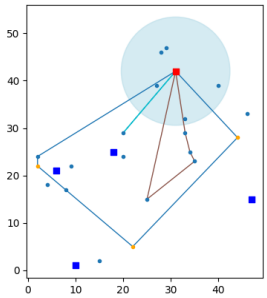
OptVax1
$C_{km} = 1$



OptVax1
$C_{km} = 12$



OptVax2 LP
-



Local
search

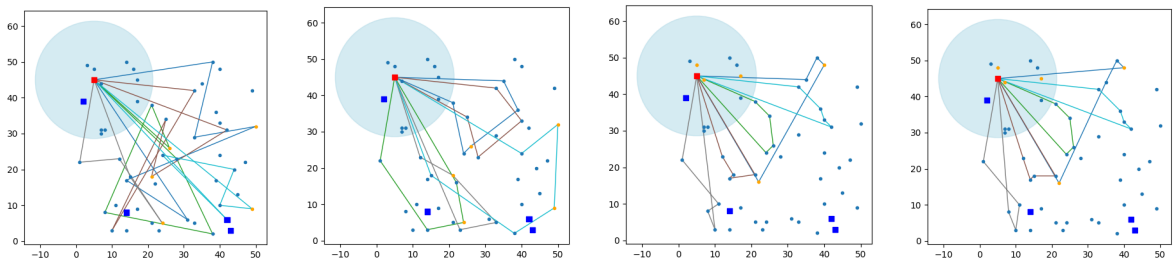
For this example, the optimization by OptVax1 gives a result of 216 vaccinated people, while the Greedy approach gives a result of 207 vaccinated people. Local search does not improve the routes for this instance.

The optimization by OptVax2 does not change the optimal value obtained by OptVax1. The interest of OptVax2 implementation is mostly related to CPU time, as it will be discussed later.

As it can be seen by the visualization and as we knew, OptVax2_LP does not give a feasible solution but an upper bound for this problem.

A last thing that is important to discuss is the difference between the two first pictures. In the first one, we can see that the routes the solver found seem not the best one. This is due to the fact that it has enough budget to permit it, and thus that the tight constraint is something else. By increasing the cost of travelling ($C_{km} = 12$) we could force the budget to be more restrictive and thus leading to better routes to save the more money as possible.

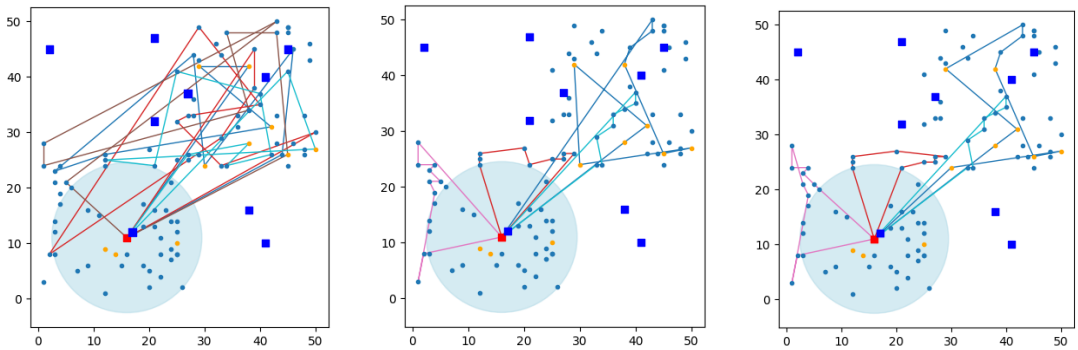
6.1.2 coord50-5-1_processed.dat



OptVax1 $C_{km} = 1$ | OptVax1 $C_{km} = 8$ | Greedy approach | Local search

For this example, the optimization by OptVax1 gives a result of 497 vaccinated people, while the Greedy approach and local approach gives a result of 457 vaccinated people. The local search algorithm improves the routes by reducing distances but not enough to allow the latest road to reach new locations.

6.1.3 coord100-10-2b_processed.dat



OptVax1 $C_{km} = 1$ | Greedy approach | Local search

For this example, the optimization by OptVax1 gives a result of 1107 vaccinated people, while the Greedy approach and local approach gives a result of 1103 vaccinated people. The local search algorithm improves the routes by reducing distances but not enough to allow the latest road to reach new locations.

Increasing the value C_{km} also increases the complexity and the running time. For this more complex example , we stay at $C_{km} = 1$. Even if it doesn't give the prettiest of visualizations

6.2 Numerical experiments

We begin by giving our results. Unfortunately, we was not able to compute a solution for instances of size 200, even not a feasible result. Also, for some instances of size 100 the solver couldn't find a optimal solution before the time out was reached, this one was fixed at $30min$. The later implies that the solution for two different models may differ.

Greedy		**Local Search**		**OptVax1**		**OptVax2**		**OptVax2 - LP**	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
256	0.0	256	0.016	265	1.25	265	1.25	265	2.84375
308	0.0	308	0.0	308	0.765625	308	0.765625	308	1.15625
207	0.0	207	0.0	216	1.125	216	1.3125	216	2.46875
302	0.0	302	0.0	302	0.640625	302	0.703125	302	2.09375

Table 1 – Results for instances of size 20.

Greedy		**Local Search**		**OptVax1**		**OptVax2**		**OptVax2 - LP**	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
457	0.0	457	0.0	497	36.671875	497	79.84375	497	6.71875
530	0.0	530	0.0	565	414.296875	537	time out	565	16.53125
482	0.0	482	0.0	510	20.71875	510	37.4375	519	10.9375
602	0.0	602	0.0	616	9.578125	616	9.3125	616	20.734375
683	0.0	683	0.0	686	11.8125	686	10.5	686	12.84375
571	0.0	571	0.0	581	4.8125	581	5.75	640	11.140625
460	0.0	460	0.0	504	15.046875	504	25.0	504	6.203125
564	0.0	564	0.0	574	14.4375	574	11.84375	578	20.34375

Table 2 – Results for instances of size 50.

Greedy		Local Search		OptVax1		OptVax2		OptVax2 - LP	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
721	0.0	721	0.0	815	time out	815	time out	851	37.25
948	0.0	948	0.016	962	1685.8281	962	1471.5469	962	532.8281
974	0.0	974	0.0	1014	121.6875	1014	246.1094	1045	36.875
1225	0.016	1225	0.0	1264	271.8125	1264	489.875	1340	36.6406
709	0.0	709	0.016	738	301.6406	738	312.4063	795	37.3906
823	0.0	823	0.016	838	time out	831	time out	1004	185.40625
796	0.016	796	0.016	765	time out	800	time out	844	319.9688
840	0.016	840	0.016	844	405.7813	844	365.75	844	802.6563
1063	0.0	1063	0.0	1063	132.875	1063	457.5	1063	218.6719
1103	0.0	1103	0.016	1107	294.625	1107	367.29688	1107	541.125
833	0.016	833	0.0	837	463.5156	837	1937.1719	837	331.0156
831	0.016	831	0.0	849	282.7031	849	444.9219	889	933.2656

TABLE 3 – Results for instances of size 100.

6.2.1 Greedy Algorithm *VS* Greedy + Local Search

First we compare the Greedy Algorithm with the Local Search. One notice few differences between both heuristic. They always give the same solution since the improvement of the Greedy Algorithm by the Local Search couldn't be exploit to add a locality to one path. There is only the CPU time that differs, but not significantly.

6.2.2 OptVax1 *VS* OptVax2

If we ignore the instances where the solver time out, we notice that the solution is always the same, meaning that the model is invariant. In contrast, the CPU time may differ between both model. Surprisingly, the valid constraint we added in the OptVax2 model seems to slow down the solver most of the time instead of helping it. We even have a time out for this one while OptVax1 find an optimal solution.

Using a test statistic on a given metric : $\sigma'_{l,k} = \frac{\tilde{z}_{l,k}}{\tau_{l,k}}$, we chose the *Wilcoxon-Mann-Whitney* one, we get a p-value of 0.672, thanks to scipy.stat, meaning that both distributions differ slightly indicating a robust result.

6.2.3 OptVax1 *VS* OptVax2 with warm-start

We give here the results for the methods run with a warm start :

The warm start does not seem to improve the CPU time but even increase it a little bit, which is surprising since it should help the solver when it is looking for a first feasible solution. However, in other optimization problem, a warm start is sometimes necessary.

OptVax1		OptVax2		OptVax2 - LP	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
265	1.296875	265	1.203125	265	2.765625
308	0.765625	308	0.78125	308	1.1875
216	1.109375	216	1.296875	216	2.453125
302	0.609375	302	0.703125	302	1.953125

TABLE 4 – Results for instances of size 20 with OptVax methods and a warm start.

OptVax1		OptVax2		OptVax2 - LP	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
497	36.75	497	83.0	497	7.375
565	410.59375	537	844.671875	565	17.3125
510	21.296875	510	40.265625	519	11.015625
616	9.578125	616	9.46875	616	24.015625
686	12.8125	686	11.6875	686	13.09375
581	4.9375	581	5.9375	640	11.140625
504	16.203125	504	26.6875	504	6.984375
574	15.296875	574	12.640625	578	22.09375

TABLE 5 – Results for instances of size 50 with OptVax methods and a warm start.

Moreover, if we reiterate the test statistic already use in the later, we find a p-value of 0.8 this time, which still indicates a great robustness.

6.2.4 OptVax2-LP

For the OptVax2-LP model, we have very interesting results. Indeed, a lot of instances have the same optimal value for the relax problem and the others. Reassuring us that the solution is optimal for the other model since it is easier to compute the solution of a non-integer problem. For all instances were the optimal solution found differ from the relax problem, it always do slightly. Also, the CPU time for solving a relax problem is much smaller than others (as we have just said, it is easier to compute so it is not surprising).

6.2.5 General comparison

Now, we will compare each model together. We first notice that unexpectedly, the heuristic such as the Greedy Algorithm or the Local Search are not too bad, they are not that far from the optimal solution. It is much more surprising since the solution given by the Local Search do not thus help the solver. To resume, we could see in the results we get that heuristic methods are much more faster than those provided by Gurobi, but even

if the solution found is not that bad, Gurobi often find the optimal solution which is our ultimate goal. Finally, if we had to chose a model for solving our problem, we will take the OptVax1 model, since the OptVax2 bring no value and can be slower. Furthermore, if one is satisfy by a quite good lower bound, the Local Search would be the best choice, and for an upper bound, choosing the OptVax2-LP model would be a good idea.

Even if our first model OptVax1 seems to be the best here, this project has familiarized us with other models which can be very useful in other contexts. Valid constraints (for OptVax2) and warm start could be very help full sometimes if it permits the solver to considerably reduce the feasible domain or to start from a solution very close to the optimal solution.

PART 2 : Approximating the optimal solution

7 Primal bound with an advanced problem specific heuristic

In this section we will present our Tabu Search's implementation. For this, we built a new class called 'Solution' that takes as argument the VC chosen for this solution as well as the routes, it also needs the instance of the problem (store as an object). This class has several useful method such as 'IsFeasible( )', 'evaluate( )' or a way to compare two equal solutions. The Solution object is thus use for applying our movements, which we describe now.

7.1 Movements

For each type of movement, we built a class that represent them. Those classes implements several useful method as 'giveNewSolution( )' that return a new solution with the movement applied (if it is feasible), or 'evaluate( )' and 'IsFeasible( )' which aim to optimize the same method for the Solution object, by using the evaluation or the feasibility of the initial solution. They also include a function that compute partially the movement. We now describe one by one the latter.

- **Merge Movement** (initial solution, (route1, route2), (a, b)) :
  This movement merges the two given routes. This is done by cutting the two routes in half at locations a and b (include and not include respectively) and collapsing them : the first part of route1 $[-, a]$ with the first of route2 $]b, -[$ and, the last part of route1 $]a, -]$ with the last of route2 $[b, -]$

- **Replace Movement** (initial solution, i, location) :
  This movement remove the given location and replace it by the best allowed location, i.e. the location that maximize the number of people vaccinated and which is feasible to take. It is possible to keep the location if it is the best. The argument i is the position of the location's route in the set of all routes.

- **Add Movement** (initial solution, i) :
  This movement tries to add a new location in the route i, by checking each non covered locality and each position possible in the given route.

- **Two_opt Movement** (initial solution, i) :
  This movement is the movement applied in the previous local search. What we do is thus swapping two locations of the route i and keeping the movement if the solution minimize the distance travelled. The swap is compute for each pair of location in the route.

Note that our implementation apply the tabu search on each VC, which are considered as fixed, with the aim to take the best one. Thus, we wont consider movement that will change the VC of the solution.

7.2 Tabu Search

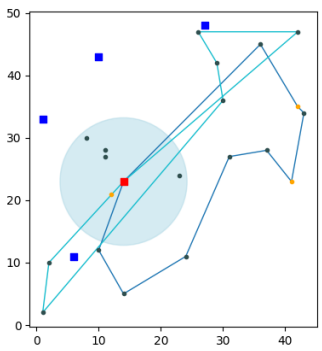
7.2.1 Algorithm

The tabu search is an heuristic that visits the neighbourhood of a solution, takes the best it has found and store as 'tabu' solutions those which were not interesting. Then it starts again this three steps until it reaches a local optimal solution or the maximum of iteration is achieved. In our case we decide that 'the best solution' is the solution that maximize the number of people vaccinated, if there is no solution which improved this objective we look for the one that minimize the needed budget (or maximize the remaining budget). When no solution is better than the current, we stop the algorithm. Finally, we add in the tabu list the solution that decrease the objective function and all previous chosen solutions so as not to go back.

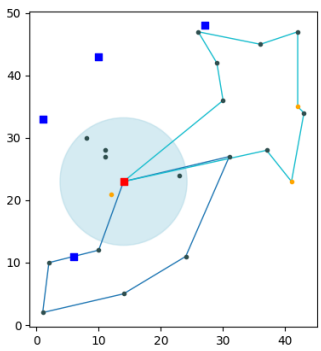
There are two parameters we have to choose to run the tabu search : the number of iterations and the size of the tabu list. Surprisingly, the number of iterations needed to reach the local optimal value is not that huge. Indeed, for the instances with 50 localities, only about fifteen iterations is enough. Thus it is not useful to pose the number of iterations up to 1000 or much more, hopping to have a better solution. In contrast, it may have a lot of solution that will be in the tabu list, for this parameter we can consider to choose a bigger length. However, this won't affect the result that much, just maybe the running time (even not significantly), since we won't look after bad solutions.

7.2.2 Example

Here is an example of the application of the tabu search on the instance 'coord20-5-2b_processed.dat'.



Greedy approach

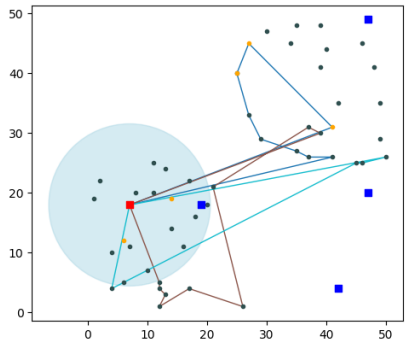


Tabu search amelioration

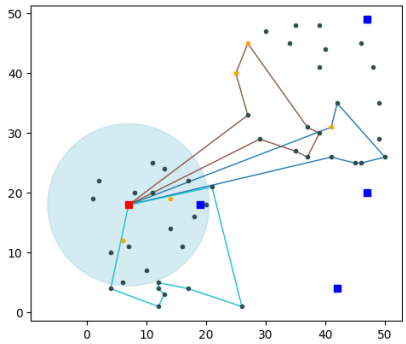
To have this result, we did the following series of movements (route 0 is the dark blue lines and the 1 is the light blue) :

1. Two_opt on the route 1
2. Merge the two routes 1 and 0
3. Two_opt on the route 0
4. Two_opt on the route 1
5. Two_opt on the route 0
6. Two_opt on the route 0
7. Two_opt on the route 1
8. Two_opt on the route 1

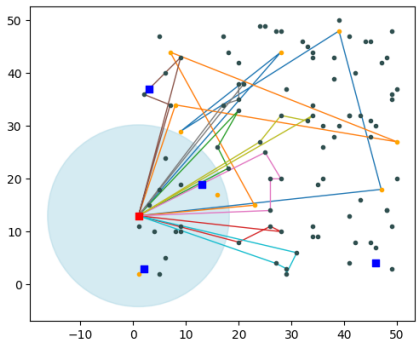
Here is two more examples on bigger instances : 'coord50-5-2b_processed.dat' and 'coord100-5-1_processed.dat'



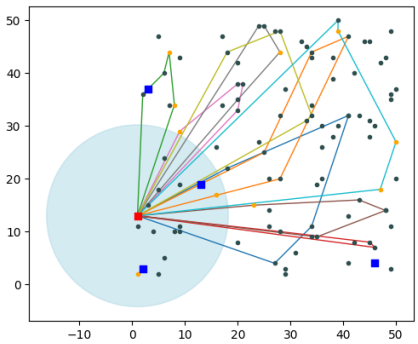
Greedy approach



Tabu search amelioration



Greedy approach



Tabu search amelioration

7.2.3 Comparison

What is interesting is now to compare the solution obtained and moreover, the CPU time the tabu search takes. Here is the three tables for instances of size $20, 50$ and 100 with the Greedy approach, the improved solution by the tabu search and finally OptVax1, which is suppose to have the optimal value (if not time out).

Greedy		Tabu Search		OptVax1	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
256	0.0	256	0.05	265	1.25
308	0.0	308	0.2618	308	0.7656
207	0.0	207	0.028	216	1.125
302	0.0	302	0.1339	302	0.6406

TABLE 6 – Results for instances of size 20.

On the instances of size 20, there is no improvement of the solution, but as we could see in the past example, the routes are better. This justify the added time.

Greedy		Tabu Search		OptVax1	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
457	0.0	486	1.021’	497	36.6719
530	0.0	530	1.5051	565	414.2969
482	0.0	499	2.1117	510	20.7188
602	0.0	616	1.7209	616	9.5781
683	0.0	683	0.8835	686	11.8125
571	0.0	581	2.4705	581	4.8125
460	0.0	498	1.9578	504	15.0469
564	0.0	564	2.2156	574	14.4375

TABLE 7 – Results for instances of size 50.

On instance of size 50 we begin to see difference between the greedy and the tabu search approach. The greedy solution is sometimes already in a local optimal solution but in other the tabu search can improve it until reach the optimal solution with almost 10 times less time (4-th instance).

Greedy		Tabu Search		OptVax1	
Sol.	**CPU**	**Sol.**	**CPU**	**Sol.**	**CPU**
721	0.0	791	11.5728	815	time out
948	0.0	962	40.8977	962	1685.8281
974	0.0	1010	2.9812	1014	121.6875
1225	0.016	1258	9.735	1264	271.8125
709	0.0	728	3.5798	738	301.6406
823	0.0	835	9.2043	838	time out
796	0.016	796	14.8888	765	time out
840	0.016	840	14.7789	844	405.7813
1063	0.0	1063	7.8202	1063	132.875
1103	0.0	1103	13.8604	1107	294.625
833	0.016	833	10.9502	837	463.5156
831	0.016	848	18.8044	849	282.7031

TABLE 8 – Results for instances of size 100.

We have very interesting results here, they bring out the weaknesses and strengths of the tabu search. Indeed, again we can see that the tabu search don't improve the solution given by the greedy algorithm. But in the first case for example, this heuristic allows us to vaccinate 70 more people in a challenging time.

8 Dual bound via Lagrangian Duality

8.1 Complicating constraints

8.1.1 Constraints Similar to a Knapsack Problem

- Budget constraint : **the total cost of choosing vaccination centers, mobilizing MMTs, and covering travel distances must stay within the fixed budget, similar to limiting the total weight in a knapsack.**

$$\sum_{i \in I} x_i f_i + \sum_{m \in M} v_m p + \sum_{m \in M} \sum_{k \in I \cup J} \sum_{j \in I \cup J} d_{ij} w_{mjk} \leq B$$

- Each route must respect the capacity constraint : **In this analogy, the population q_j represents the "weight," and Q represents the total "weight" that must not be exceeded, similar to a knapsack problem.**

$$\forall m \in M \qquad \sum_{j \in J} t_{mj} q_j \leq v_m Q.$$

- The capacity for the VC is only for specifically conditioned doses delivered by MMTs : **As before, the population q_j represents the "weight," and C_i represents the total "weight"**

$$\forall i \in I \qquad x_i \sum_{m \in M} \sum_{j \in J} t_{mj} q_j \leq C_i.$$

8.1.2 Constraints Similar to an Assignment Problem

- One and only one vaccination center must be opened : **Corresponds to a single assignment, where you must choose exactly one "object" (in this case, the VC).**

$$\sum_{i \in I} x_i = 1.$$

- A locality can be either ; not visited, visited at most once by a MMT or within the range of an opened VC : **Assign each locality to an MMT or VC, or leave it uncovered**

$$\forall j \in J \qquad \sum_{m \in M} t_{mj} + \sum_{i \in I} y_{ij} \leq 1.$$

- Each route corresponds to a mobilized MMT : **Each location is assigned to a single MMT for entry and exit**

$$\forall j \in J \qquad \sum_{m \in M} \sum_{\substack{k \in I \cup J \\ k \neq j}} w_{mjk} \leq 1$$

$$\forall k \in J \qquad \sum_{m \in M} \sum_{\substack{j \in I \cup J \\ j \neq k}} w_{mjk} \leq 1.$$

- All prioritized localities have to be covered : **Assign each priority locality to an MMT or VC**

$$\forall j \in \hat{J} \qquad \sum_{m \in M} t_{mj} + \sum_{i \in I} y_{ij} = 1.$$

8.1.3 Constraints Similar to a Shortest Path Problem

- The sum of all MMT starting from a location is exactly the sum of MMT arriving to this location : **similar to a shortest path problem : the flow conservation property is satisfied**

$$\forall m \in M, \quad \forall k \in I \cup J \qquad \sum_{\substack{j \in I \cup J \\ j \neq k}} w_{mjk} = \sum_{\substack{j \in I \cup J \\ j \neq k}} w_{mkj}.$$

- The starting point and the end point are the same for each route and turns out to be the chosen VC : **similar to a shortest path problem with a cyclical route, where each path begins and ends at the same source node (Here, the VC).**

$$\forall i \in I \qquad \sum_{m \in M} \sum_{j \in I \cup J} w_{mij} = x_i \sum_{m \in M} v_m$$

$$\forall i \in I \qquad \sum_{m \in M} \sum_{j \in I \cup J} w_{mji} = x_i \sum_{m \in M} v_m.$$

8.1.4 Reformulation of the objective function

Assuming that the previously mentioned constraints form the subset of easy constraints, it remains to deal with the constraint of the elimination of sub-turns.

$$\forall m \in M, \quad \forall j,k \in J, k \neq j \qquad w_{mjk} + u_{mj} \leq u_{mk} + (|I| + |J|)(1 - w_{mjk}).$$

We thus have the following objective function for the Lagrangian dual :

$$\text{Max} \sum_{j \in J} q_j \left(\sum_{m \in M} t_{mj} + \sum_{i \in I} y_{ij} \right) + \sum_{m \in M} \sum_{k \in J} \sum_{j \in J} \lambda_{mkj} \left(u_{mj} + (|I| + |J|)(1 - w_{mkj}) - w_{mkj} - u_{mk} \right)$$

Avec $\lambda_{mkj} \geq 0 \quad \forall m \in M, \quad \forall j,k \in J$

8.2 Sub-gradient Descent

We are asked to implement a sub-gradient method so as to minimize the Lagrangian Dual objective. The subgradient algorithm works essentially like gradient descent, but replaces the gradient with a subgradient. It iteratively updates the dual variables in the direction of the subgradient of the Lagrangian to improve the dual objective. We want to minimize the objective function as a function of lambda.

$$\min_{\lambda} z(\lambda) = \min_{\lambda} \max_{x \in X} c^T x + \lambda^T (d - Dx)$$

The goal is therefore to gradually "drive lambda to 0" when the constraint is respected (and therefore brings positive terms) but to always penalize the constraints that are not respected.

From what we've seen in class $(d - Dx(\lambda))$ is a subgradient of $z(\lambda)$. We use the following subgradient algorithm.

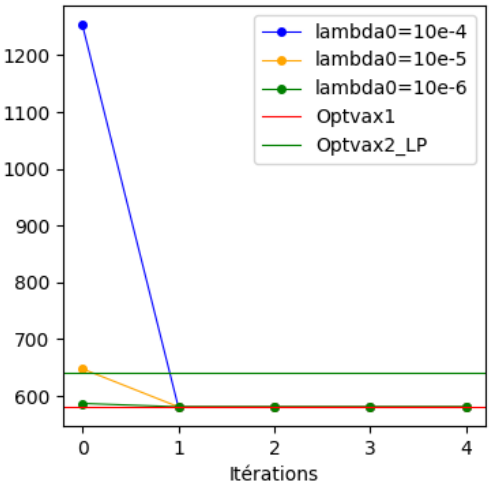
Choose initial λ_0 ;
for *k = 0, 1, 2, . . .* **do**
  $\lambda_{k+1} = \max(\lambda_k - h_k(d - Dx(\lambda_k), 0)$
end

Where h_k must be chosen appropriately. Typically, h_k is selected to ensure that the method converges to an optimal dual solution. However, if the steps remain large for too long, the method may oscillate. And on the contrary, if it decrease too quickly, it will take longer to reach the optimum. Here we choose an h_k in the form $\frac{a}{k+b}$ whith $a, b > 0$. We will detail the choice of a and b just after.

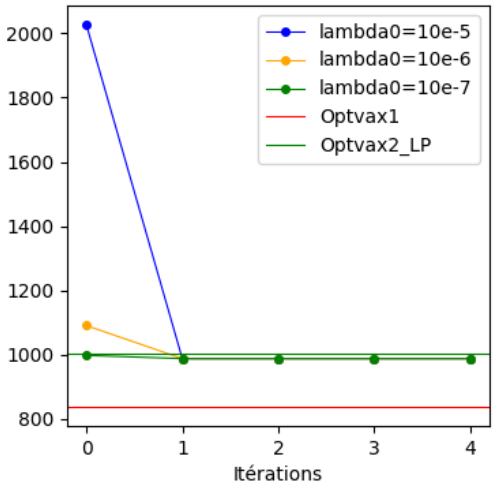
8.3 Step size strategy

8.3.1 Influence of λ_0

To study the influence of λ_0, we fix in this part $h_k = \frac{a}{k+b}$ with $a = 0,01$ and $b = 100$.



coord50-5-2BIS_processed



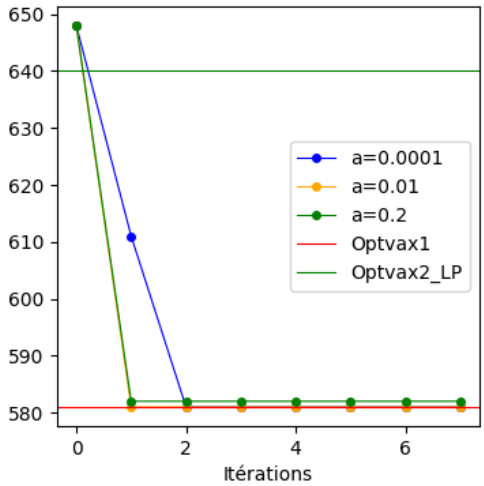
coord100-5-3b_processed

Influence of λ_0 on convergence to the Dual Upper Bound

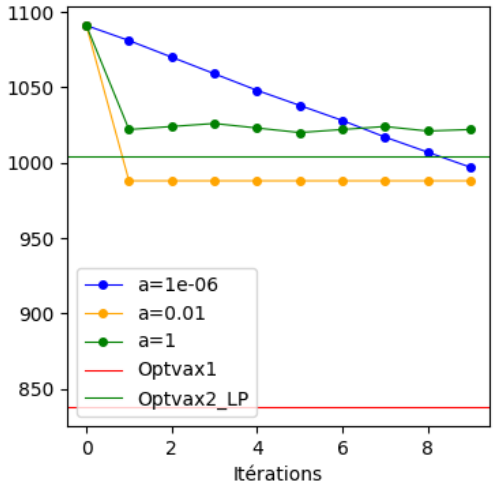
Thus, lambda does not influence the convergence but only the first results. For visualization purposes, we will choose a λ_0 adapted according to the instances. The more roads there are, as in large instances, the more the constraints of the sub-tours bring to the first iteration an important term in the sum to be maximized. We will then start with a smaller λ_0.

8.3.2 Influence of a in $h_k = \frac{a}{k+b}$

To study the influence of a in $h_k = \frac{a}{k+b}$, we fix in this part $b = 100$.



coord50-5-2BIS_processed



coord100-5-3b_processed

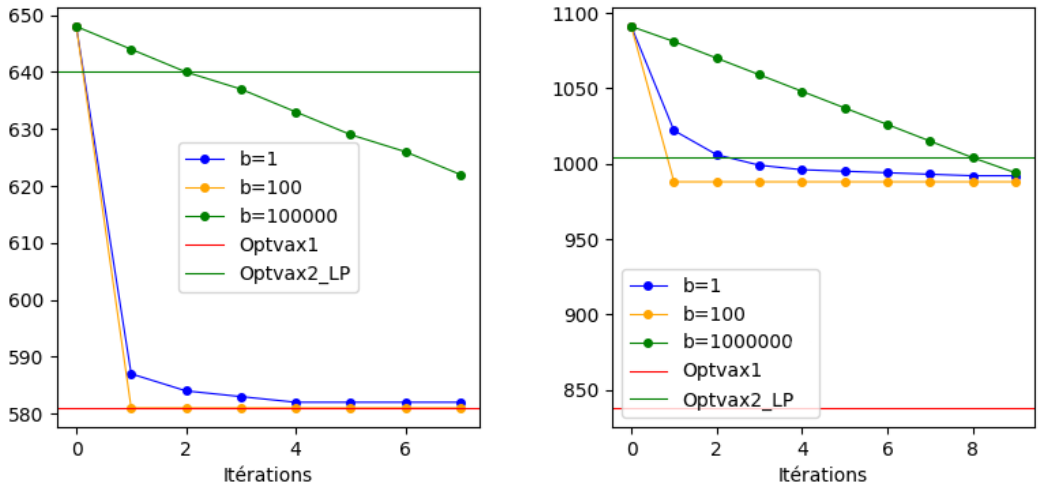
Influence of a in the expression of h_k on convergence to the Dual Upper Bound

The choice of a influences the convergence. As observed, a a that is too small or too

large is not optimal. However, this behaviour varies across instances : $a = 1e-4$ is suitable for large instances, but is not optimal for smaller ones. Thus, we opt for $a = 0.01$ for all instances in the following, which gives fast convergence in all cases.

8.3.3 Influence of b in $h_k = \frac{a}{k+b}$

To study the influence of b in $h_k = \frac{a}{k+b}$, we fix in this part $a = 0.01$.



coord50-5-2BIS_processed coord100-5-3b_processed

Influence of b in the expression of h_k on convergence to the Dual Upper Bound

As for a, the choice of b influences the convergence. A b that is too small or too large is not optimal. According to the results, we therefore opt for $b = 100$

9 Numerical Experiments

As explained, we set $a = 0.01$, $b = 100$ to perform the various tests.

It is not particularly relevant to directly compare the execution time of the Lagrangian Duality method between the different methods since it depends a lot on the choice of h_k and the number of iterations k.

However, we can note that with a relevant h_k we can find the optimal bound in less than 8 iterations. Each iteration, however, requires running Optvax1 dual Lagrangian to have at each time the maximum with λ fixed, $\max_{x \in X} c^T x + \lambda^T (d - Dx)$.

Greedy	Local Search	OptVax1	Lag dual	OptVax2 - LP
721	721	815	815	851
948	948	962	962	962
974	974	1014	1014	1045
1225	1225	1264	1265	1340
709	709	738	739	795
823	823	838	989	1004
796	796	765	844	844
840	840	844	844	844
1063	1063	1063	1063	1063
1103	1103	1107	1107	1107
833	833	837	837	837
831	831	849	849	889

TABLE 9 – Results for instances of size 100.

For instances 20 and 50, the dual Lagrangian provides us with an ideal upper bound, since we find the same result as for Optvax1. But on these small instances, this method is not the most efficient since it can run unnecessarily (and therefore it takes longer than OptVax1). Nevertheless, it turns out to be very efficient for large instances as it almost always provides an upper bound that matches the solution found with Optvax1 but in much less time.

10 Conclusion

To conclude, the most promising method we would advise another island to use so as to find an effective immunization campaign is the Tabu Search. It seems to be a very good compromise between having a great solution and the rapidity. It is not too far from the optimum and rarely exceed 20 CPU. One other advantage is that the solution is certainly feasible. Unlike dual Lagrangian, which also give a quick result but do not always give a feasible solution. This may be important in this example if the vaccination campaign is to be implemented quickly.